

Q1) What libraries are used in Python for data science? ----- Common Python libraries used in Data Science are: 1) NumPy – For numerical computations and arrays. - 2) Pandas – For data manipulation and analysis. - 3) Matplotlib – For data visualization (graphs, charts). - 4) Seaborn – For advanced and attractive visualizations. - 5) Scikit-learn – For machine learning algorithms. - 6) TensorFlow / PyTorch – For deep learning and AI models. - 7) Statsmodels – For statistical analysis. - 8) SciPy – For scientific and mathematical operations.

Q2) Explain Cross Validation ----- Cross Validation is a technique to test the model's performance on multiple data splits.

☞ K-Fold Cross Validation: * Dataset divided into K parts (folds). * Model trained on (K-1) parts and tested on 1 part. * Repeated K times and average accuracy is taken. | It ensures more reliable results than a single train/test split.

Q3) What is data science life process? (or stages in data science) ----- The Data Science Life Cycle includes the following stages: 1) Data Collection – Gathering data from various sources. 2) Data Cleaning – Removing errors, missing or duplicate data. 3) Data Exploration / Analysis – Understanding and visualizing data. 4) Data Modeling – Building machine learning models. 5) Model Evaluation – Checking the accuracy of the model. 6) Deployment – Using the model in real applications. 7) Monitoring & Maintenance – Improving and updating the model.

Q4) What are applications of data science in the real world? ----- 1. Healthcare – Disease prediction, medical image analysis. 2. Finance – Fraud detection, credit scoring. 3. E-commerce – Product recommendation (like Amazon). 4. Transportation – Route optimization, self-driving cars. 5. Social Media – Trend analysis, sentiment detection. 6. Education – Student performance analysis. 7. Sports – Player performance and match predictions.

Q5) What is data analytics and its three types? ----- Data Analytics means examining raw data to find useful insights and support decision-making. **Types of Data Analytics:** 1) Descriptive Analytics – What happened?

Example: Monthly sales report. 2) Predictive Analytics – What will happen?

Example: Predicting next month's sales. 3) Prescriptive Analytics – What should we do?

Example: Recommending strategies to improve sales.

Q6) What is Data Collection and its Types? ----- Data Collection means gathering information from different sources to analyze and make decisions. **Types of Data Collection:** 1) Primary Data – Collected first-hand by the researcher. *Examples:* Surveys, interviews, experiments. 2) Secondary Data – Collected by someone else earlier.

Examples: Government reports, company databases, websites.

Q7) How to Handle Missing Values and Outliers? ----- Missing Values Handling: 1) Remove missing data – If only a few records are missing. 2) Fill with mean/median/mode – Replace missing values with average or most common value. 3) Use interpolation or model prediction – Estimate missing data using ML models.

Outliers Handling: ----- 1) Remove outliers – If they are errors or invalid data. 2) Cap or limit values – Replace extreme values with a maximum/minimum limit. 3) Use transformation – Apply log or square root to reduce outlier effect.

Q8) What is Data Transformation, Feature Scaling, and Encoding Categorical Data? ----- **Data Transformation:** | Changing data into a suitable format for analysis (like normalization, aggregation, or converting data types).

Feature Scaling: Bringing all numerical values to the same scale so that no variable dominates others. | **Methods:** | 1) Min-Max Scaling – Converts values between 0 and 1. 2) Standardization (Z-score) – Converts data to have mean = 0, std = 1

Encoding Categorical Data: Converting text (categories) into numbers. | **Methods:** Label Encoding – Assigns each category a number. | One-Hot Encoding – Creates separate columns for each category.

Q9) What is Data Integration? ----- Data Integration means combining data from multiple sources into one unified view. | **Example:** Merging sales data from different branches or systems. | It helps in complete and consistent analysis.

Q10) What is Descriptive Statistics? ----- Descriptive Statistics summarizes and describes main features of a dataset. It includes: | 1) Measures of Central Tendency (mean, median, mode) 2) Measures of Dispersion (range, variance, standard deviation). 3) Data visualization (charts, tables)

Q11) What is Central Tendency and Dispersion? ----- Central Tendency: Shows the center or typical value of data. | Types: Mean, Median, Mode.

Dispersion: Shows how spread out or varied the data is. Types: Range, Variance, Standard Deviation

Example: If all students score around 70, dispersion is low; if scores vary from 30–100, dispersion is high.

Q12) What is Distribution Analysis? ----- Distribution Analysis studies how data values are spread or distributed. It shows patterns such as: | * Normal Distribution – Bell-shaped curve (most values around mean). * Skewed Distribution – More values on one side of mean. | It helps understand data behavior and detect outliers.

Q13) What is Exploratory Data Analysis (EDA)? ----- EDA is the process of exploring data to discover patterns, trends, and relationships before modeling. **Steps include:** 1) Data cleaning 2) Summary statistics 3) Visualization (histogram, box plot, scatter plot) 4) Correlation analysis **Goal:** To understand the dataset and get insights for further analysis.

Q14) What is Regression? And its Types? ----- Regression is a supervised machine learning technique used to predict a continuous (numerical) output based on input data. **Example:** Predicting house prices, temperature, or sales. **Types of Regression:** 1) Linear Regression – Predicts output using a straight-line relationship between input and output. 2) Multiple Linear Regression – Uses multiple features for prediction. 3) Polynomial Regression – Fits data with a curve (non-linear). 4) Logistic Regression – Used for classification (yes/no outcomes). 5) Ridge / Lasso Regression – Used to reduce overfitting by adding regularization.

Q15) What is Classification and its Types? ----- Classification is a supervised learning technique used to predict categorical outcomes (like Yes/No, Spam/Not Spam). **Example:** Email spam detection, disease prediction. **Types of Classification:** 1) Binary Classification – Two categories (e.g., Male/Female). 2) Multi-Class Classification – More than two categories (e.g., Type of fruit). 3) Multi-Label Classification – Each item can belong to multiple classes (e.g., tagging an image as “animal” and “pet”).

Q16) What is Splitting Data and Training the Model? ----- **Splitting Data:** Dividing the dataset into parts — usually Training Set and Testing Set. | **Example:** 80% training, 20% testing. * from sklearn.model_selection import train_test_split | * X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2) | **Training the Model:** The model learns the relationship between inputs and outputs using the training data.

Q17) What are Model Evaluation Metrics? ----- Model Evaluation Metrics are used to measure how well a machine learning model performs. **Examples:** * For Classification: Accuracy, Precision, Recall, F1 Score, AUC-ROC * For Regression: MSE, RMSE, MAE, R² Score

Q18) What is Confusion Matrix? Explain with Example ----- A Confusion Matrix shows how well a classification model performs by comparing actual and predicted values.

Example:

If a spam detector predicts:

- TP = 80 (correct spam)
- TN = 90 (correct not spam)
- FP = 10 (wrongly predicted spam)
- FN = 5 (missed spam) | → This helps calculate Accuracy, Precision, Recall, etc.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Q19) Explain Classification Metrics ----- 1) Accuracy: Fraction of correct predictions. | $\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$ 2) Precision: | Out of all predicted positives, how many are correct. | $\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$ 3) Recall (Sensitivity): Out of all actual positives, how many were predicted correctly. | $\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$ 4) F1 Score: Harmonic mean of Precision and Recall. | $\text{F1} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$ 5) AUC-ROC Curve: * ROC (Receiver Operating Characteristic) curve plots True Positive Rate vs False Positive Rate. * AUC (Area Under Curve) shows overall model performance (closer to 1 = better).

Q20) Explain Regression Metrics ----- 1) MSE (Mean Squared Error): Average of squared difference between actual and predicted values. | $\text{MSE} = \frac{1}{n} \sum (y_{\text{actual}} - y_{\text{pred}})^2$ 2) RMSE (Root Mean Squared Error): Square root of MSE — gives error in same units as target variable. | $\text{RMSE} = \sqrt{\text{MSE}}$ 3) MAE (Mean Absolute Error): Average of absolute differences between actual and predicted values. | $\text{MAE} = \frac{1}{n} \sum |y_{\text{actual}} - y_{\text{pred}}|$ **Difference:** 1) MAE – Easier to interpret (average error). 2) MSE – Penalizes large errors more. 3) RMSE – Most commonly used for performance measure.

Q21) Explain Indexing and Slicing in NumPy Array ----- **Indexing:** | Accessing specific elements from a NumPy array using their position (index). **Example:** | import numpy as np | arr = np.array([10, 20, 30, 40]) | print(arr[2]) # Output: 30 | **Slicing:** Accessing a range of elements from an array using start:end:step. 🖱️ **Example:** print(arr[1:3]) # Output: [20 30] | print(arr[::2]) # Output: [10 30]

Q22) What are the Basic Mathematical and Statistical Operations in NumPy Array? ----- **Mathematical Operations:** 1) +, -, *, / – Element-wise operations 2) np.sum(arr) – Sum of elements. 3) np.mean(arr) – Average value 4) np.max(arr) / np.min(arr) – Max and Min values 5) np.sqrt(arr) – Square root 6) np.exp(arr) – Exponential. **Statistical Operations:** 1) np.median(arr) – Median value 2) np.std(arr) – Standard deviation 3) np.var(arr) – Variance

Q24) What is Data Visualization? Explain Different Types of Plots ----- Data Visualization is the graphical representation of data to understand patterns and trends. ---|--- **Common Plot Types:** **1)** Line Plot – Shows trends over time. (plt.plot()) **2)** Bar Chart – Compares categories. (plt.bar()) **3)** Histogram – Shows frequency distribution. (plt.hist()) **4)** Scatter Plot – Shows relation between two variables. (plt.scatter()) **5)** Pie Chart – Shows parts of a whole. (plt.pie()) **6)** Box Plot – Shows data spread and outliers. (plt.boxplot())

Q25) What is a Pivot Table? ----- A Pivot Table summarizes and reorganizes data to show useful patterns. | **In pandas:** `pd.pivot_table(data, values='Sales', index='Region', columns='Year', aggfunc='sum')` | It helps in analyzing data efficiently, similar to Excel pivot tables.

Q26) What is Map() and Replace() Function Used For? ----- `map()`: Applies a function or mapping to each element in a Series. | **Example:** `s = pd.Series([1,2,3]) | s.map({'1':'A',2:'B',3:'C'})` | `replace()`: Replaces specific values with new ones. **Example:** `s.replace(2, 200)`

Q27) What is plt.legend() Function Used For? ----- The `plt.legend()` function in Matplotlib displays labels for multiple plots on the same graph. **Example:** `| import matplotlib.pyplot as plt | plt.plot([1,2,3], label='Line A') | plt.plot([3,2,1], label='Line B') | plt.legend() # shows the legend box | plt.show()`

Q28) What is DataFrame and Series? ----- **Series:** * A 1D labeled array (like a single column). | **Example:** `import pandas as pd | s = pd.Series([10,20,30])` | **DataFrame:** * A 2D labeled data structure (like an Excel table). | **Example:** `data = {'Name':['A','B'], 'Age':[20,25]} | df = pd.DataFrame(data)`

Q29) What is Indexing and Slicing in Series? ----- **Indexing:** Access elements using labels or positions. `s = pd.Series([10,20,30], index=['a','b','c']) | print(s['b'])` # Output: 20
Slicing: Access range of data using : `| print(s[0:2])` # Output: a 10, b 20

Q30) Explain Overfitting and Underfitting

Type	Description	Example
Overfitting	Model learns training data too well, including noise → performs poorly on new data.	High training accuracy, low test accuracy
Underfitting	Model fails to learn patterns from data.	Both training & test accuracy low

👉 Goal: Find a balance between the two.

Q31) Difference between Structured and Unstructured Data

Feature	Structured Data	Unstructured Data
Format	Organized (tables, rows, columns)	Unorganized (text, images, videos)
Storage	Stored in relational databases (SQL)	Stored in data lakes, NoSQL, or files
Ease of Analysis	Easy to analyze using traditional tools	Hard to analyze, needs AI/ML tools
Example	Student database, Excel sheets	Emails, photos, YouTube videos

Q32) What is Flatten() and Ravel()?

Both convert multi-dimensional arrays into 1D arrays.

Function	Description	Returns
<code>flatten()</code>	Creates a new copy of the array in 1D	Returns a copy
<code>ravel()</code>	Returns a 1D view of the same data	Returns a view (no copy)

Example: `arr = np.array([[1,2],[3,4]]) | arr.flatten() # [1 2 3 4] | arr.ravel() # [1 2 3 4]`

33) What is Correlation? Explain its types? And which graph is used to check correlation between data?

→ **Correlation:** Measures the strength and direction of relationship between two variables.

Types of Correlation: **1)** Positive Correlation: Both variables increase together. **2)** Negative Correlation: One variable increases, the other decreases. **3)** No Correlation: No predictable relationship.

Graph used to check correlation: **1)** Scatter Plot – visually shows correlation. **2)** Heatmap – shows correlation coefficient between multiple variables.

34) What is Skewness? Explain its types? And which graph is used to check skewness? ----- **Skewness:** Measures asymmetry in the distribution of data. | **Types of Skewness:** **1)** Positive Skew (Right Skew): Tail is longer on the right. **2)** Negative Skew (Left Skew): Tail is longer on the left. **3)** Zero Skew: Symmetrical distribution (normal distribution).

Graph used to check skewness: 1) Histogram – Shows shape of data distribution. 2) Box Plot – Can indicate skew by position of median in the box.

35) What is Model Building? ----- Model Building: Process of creating a predictive model using machine learning.

Steps: 1) Understand problem & collect data. 2) Clean and preprocess data. 3) Select features and choose algorithm. 4) Split data into training and testing sets. 5) Train the model on training data. 6) Evaluate performance on test data. 7) Tune parameters and finalize model.

36) Explain what EDA is, in detail? ----- EDA (Exploratory Data Analysis): Process of analyzing datasets to summarize main characteristics, often visually. **# Purpose:** * Understand data structure and patterns. * Detect missing values and outliers. * Identify relationships between variables.

Common EDA Techniques: 1) Summary Statistics: mean, median, mode, std deviation. 2) Visualization: histogram, box plot, scatter plot, pair plot, heatmap. 3) Correlation Analysis: to check relationships between variables. 4) Handling Missing Values & Outliers

37) Explain in Detail: What is Encoding? What are the two types of encoding?

Encoding: Converting categorical/text data into numerical format for ML models.

Types of Encoding: 1) Label Encoding: Assigns unique numbers to categories. | **Example:** ["Red", "Blue", "Green"] → [0,1,2]

2) One-Hot Encoding: Creates separate binary columns for each category. | **Example:** ["Red", "Blue", "Green"] → Red:[1,0,0], Blue:[0,1,0], Green:[0,0,1]

38) What is InterQuartile Range (IQR) and how is it calculated? Explain it with an example. ----- InterQuartile Range (IQR): Measures spread of the middle 50% of data.

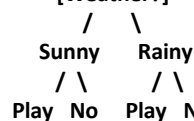
Formula: $IQR = Q3 - Q1$ | Where: | $Q1 = 25\text{th percentile}$ | $Q3 = 75\text{th percentile}$

Example: Data: [10, 12, 15, 18, 20, 22, 25] | $Q1 = 12$ | $Q3 = 22$ | $IQR = 22 - 12 = 10$

Use: | Detect outliers: | Lower bound = $Q1 - 1.5 * IQR$ | Upper bound = $Q3 + 1.5 * IQR$

39) What is Decision Tree? Explain with Diagram----- Decision Tree: * A Decision Tree is a supervised machine learning algorithm used for classification and regression. * It splits data into branches based on conditions on features, forming a tree-like structure of decisions. * Components: 1) Root Node – The starting point (first feature split). 2) Internal Nodes – Represent feature tests. 3) Leaf Nodes – Represent final output (class or value).

Example Diagram: [Weather?]



Here, the tree decides whether to play based on weather.

40) What is Clustering? Explain Types of Clustering . ----- **Clustering:** Unsupervised learning technique to group similar data points. **Types:** 1) K-Means: Divides data into K clusters based on distance to centroids. 2) Hierarchical:

Builds a tree (dendrogram) of clusters; can be agglomerative or divisive. 3) DBSCAN: Clusters based on density; detects arbitrary shapes and outliers. 4) Fuzzy Clustering: Points can belong to multiple clusters with degrees of membership. Visualization: Scatter plot or dendrogram.

41) Explain Random Forest ----- **Random Forest:** | An ensemble method using multiple decision trees for classification/regression. | **How it works:** 1) Bagging: Random subsets of data train each tree. 2) Random Features: Each

tree uses random subset of features. 3) Voting/Averaging: | Classification: majority vote | Regression: average prediction

Advantages: * High accuracy, reduces overfitting * Handles missing values and outliers * Robust and flexible

Example Use Cases: Customer churn prediction, fraud detection, disease prediction.